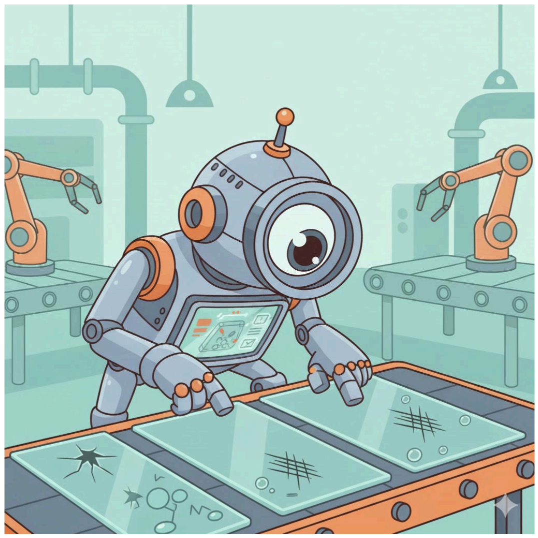
ZURICH UNIVERSITY OF APPLIED SCIENCES

DEPARTEMENT LIFE SCIENCES AND FACILITY MANAGEMENT

Glass Defect Detection: Evaluating Object Detection Models



Confidential

Projectwork 2

by

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Bachelors 2023

Submission date: 2025-12-11

Study Direction: Applied Digital Life Science

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Imprint

Recommended Citation:

Florian Vollmer (2025). *Glass Defect Detection: Evaluating Object Detection Models*

.

Keywords: Deep Learning, Object Detectors, Image Processing, Evaluation Metrics

Abstract

Write after work is written

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List of abbrevations

Potential abbrevations

1. Introduction

1.1. Context and Motivation

In the current technological landscape, the implementation of modern technologies is indispensable for ensuring precise and automated processes. Within the industrial sector, quality control represents a critical success factor. Specifically in pharmaceuticals and medical technology, the integrity of the primary packaging material (typically glass containers) is crucial for product safety.

The primary material used for medical and chemical glass containers generally consists of borosilicate glass. However, this material can exhibit or contain production-related defects, including particulate inclusions, air bubbles, or flexible fragments known as Lamellae (Foglia, Prete, and Zanda 2015).

The relevance of these quality deficiencies is highlighted by the consequences demonstrated by (Foglia, Prete, and Zanda 2015): *“In recent years, more than 20 product recalls have been associated with glass issues, and over 100 million units of drugs packaged in vials or syringes have been withdrawn from the market.”* This underscores the urgent necessity of reliable detection methods for such flaws.

The rapid advancement of Artificial Intelligence (AI) and fundamental mechanisms like Deep Learning allows for the application of so-called object detectors for the automated recognition of the aforementioned defects. The attractiveness of these AI-based image recognition methods is continuously increasing (Bhatt et al. 2021). However, as these procedures still represent a nascent field of research, comprehensive validation studies on their suitability and the achievability of the necessary performance in industrial use are lacking.

A central challenge in training robust models is the scarcity of extensive datasets (Bhatt et al. 2021). Specifically, the naturally limited availability of data from defective products, caused by the significant imbalance in favor of intact units, complicates the creation of an adequate training dataset (Bhatt et al. 2021).

This paper serves as an exploratory contribution to the evaluation of the current feasibility of AI-supported defect detection. The objective of this work is the comparison of available models on the market and the existence of suitable datasets, in order to provide a well-founded statement on the realisability of precise defect determination.

1.2. Overall Concept

The overall objective of this work was to conduct a quantitative performance comparison of two different object detection models under varying data conditions. The experimental procedure was based on the following fundamental steps:

1. **Model selection and preparation:** Two specific, pre-trained foundation models from the class of object detectors were selected and initialized for evaluation.
2. **Data management and fine-tuning:** To test the models' generalizability, they were each adapted to the specific classification task (defect detection) using two discrete datasets (dataset A and dataset B) via a fine-tuning process.
3. **Performance evaluation:** The performance of the four trained model-dataset combinations was then evaluated on a separate, previously unseen test dataset. The performance was determined based on established quantitative metrics for object detection.
4. **Comparative analysis:** Finally, the generated results were visualized and subjected to a detailed comparative analysis. The aim was to identify the robustness, strengths, and weaknesses of the model architectures examined in relation to glass defect detection.

1.3. Research Questions

Based on the challenges mentioned in **Context and Motivation**, this study tries to answer following research questions:

- How does the performance differ between the two investigated object detection models (Model X vs. Model Y), measured by Mean Average Precision (mAP), in detecting defects in the different datasets?
- Which performance benchmarks (e.g., minimum precision and recall values) must be achieved to deem the resulting reliability of the models as sufficient for deployment in real-world quality control scenarios?

1.4. State of the Art

Ganz kurz: was sind objekt detektoren, wie funktionieren sie, was ist IoU # Methodology ## Datasets and Assumptions The search for suitable data sets proved to be challenging due to the inherent imbalance in industrial production. As mentioned in the introduction, the lack of data is a key problem in the context of defect detection. For the experimental implementation, two separate data sets were selected. These data sets represent different product forms, but have the identical defect type of gas bubbles.

Dataset A (Glasjars, Roboflow)

The first dataset was an annotated dataset of glass jars. This dataset was obtained from the Roboflow platform.

- **Defined classes:** This dataset contained two primary classes: the defect class (the gas bubble itself) and the container class (the glass container as an object). The models were trained to locate both the surrounding product and the specific defect using bounding boxes.
- **Scope and division:** The dataset comprised a total of 1728 images. The division into the three essential subcategories of training set, validation set ('val') and test set ('test') was already predefined. This division was adopted for the subsequent analysis.

Figure 1: Raw example image of the dataset A

Figure 1 shows that the images were caputred in monotone coloring, they are not autorotated, meaning the angle of the object is different in every picture.

Dataset B (Glass Plates, Kaggle)

The second dataset was obtained from the Kaggle platform and was used to verify the model's performance on structurally different image data.

- **Scope and distribution:** With a total of 208 images, this dataset was significantly smaller than dataset A. Here, too, the distribution into training, validation and test sets was predefined and was adopted for the analysis.
- **Product and class definition:** In contrast to the first data set, data set B did not contain images of entire pharmaceutical containers, but rather images of a glass plate that exhibited the defect. The classification was limited exclusively to the defect class (gas bubble). This set came annotated with the respective bounding boxes and the class "DEFECT".
- **Important Feature:** A key feature of this data set was the guarantee that each image in the sample contained at least one defect. This represented a contrasting data basis to real production environments, as the natural imbalance between defective and intact samples was not represented here.

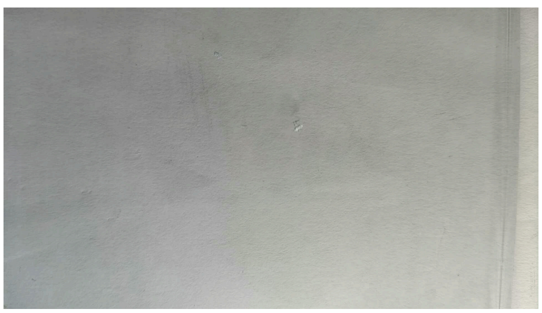

Figure 2: Example image of dataset B

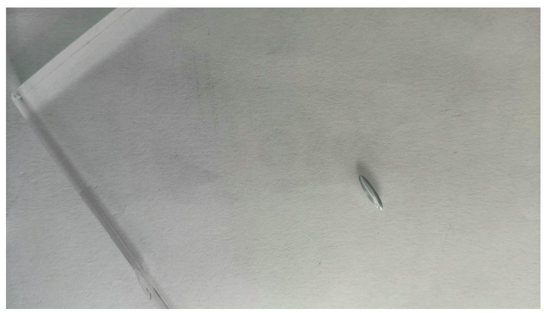

Figure 3: Example image of dataset B

As shown in (Figure 2) and Figure (Figure 3), the extent of the gas bubbles varied significantly from microlesions to large-area defects. This variation placed demands on the localisation capability of the models across a wide range of sizes.

In view of the structural and qualitative differences between the data sets, a number of hypotheses were formulated in advance.

It was assumed that data set A would be more challenging due to the different rotation angles of the images and the more complex classification. A longer training period and lower overall performance were expected here compared to data set B.

On the other hand, the small size of dataset B led to the hypothesis of low generalisability, although rapid learning was assumed on the internal test images.

1.5. Model Selection

For the comparative part of the study, the decision was focused on two of the currently best-known architectures for object detection. This selection aimed to evaluate the performance of a conventional convolutional neural network (CNN)-based approach against a transformer-based model.

Ansprechen was in den ground truth files drin steht mit superclass, für was coco steht und warum es anders ist. dass die daten von kaggle auf roboflow von yolo zu coco konvertiert wurden

- **YOLO (You Only Look Once)** The YOLO model was selected as the first reference detector. The model was developed by the software developer Ultralytics and is considered a real-time object detection and segmentation model. It is characterised by its high speed and efficient performance, making it one of the most commonly used one-stage detectors.(Ultralytics (2025))

For this task the YOLO Model Version 11 Nano was chosen.

- **RF-DETR (Roboflow Detection Transformer)** The RF-DETR model, provided by Roboflow, served as an alternative architecture. RF-DETR is a transformer-based model for object detection and instance segmentation. It represents a newer class of detectors which, unlike CNN-based architectures, does not require manual post-processing (such as non-maximum suppression).

1.6. Model Mechanics

- **YOLO (You Only Look Once)** The YOLO (You Only Look Once) model is a real-time object detection system which fundamentally differs from traditional multi-stage detectors by processing the input image in a single pass. This architectural choice classifies YOLO as a Single-Shot Detector (SSD).

This type of detector employs an algorithm that predicts the bounding box coordinates and the corresponding class probability of the detected object simultaneously and in one step. The primary beneficial consequence of this unified approach is a significant increase in processing speed. (Buhl (2024)) Consequently, the simultaneous prediction of the object's presence and its precise location accelerates the overall detection process, making it highly suitable for real-time applications.

- **RF-DETR (Roboflow Detection Transformer)** As the comparative alternative architecture, the RF-DETR (Roboflow Detection Transformer) model was selected. This model represents a real-time, state-of-the-art transformer-based architecture specifically designed for object detection and instance segmentation. (Robinson et al. 2025).

The RF-DETR architecture leverages a Vision Transformer (ViT) to effectively extract multiscale features from the input image.

Following feature extraction, the model utilizes a balanced combination of two distinct attention mechanisms:

Windowed Attention Blocks: These restrict the self-attention mechanism to a local neighbourhood, which significantly improves processing speed.

Non-Windowed Attention Blocks: These enable global interaction across features, which contributes to overall detection accuracy.

This deliberate balance helps maintain high performance while ensuring the model remains fast enough for real-time applications. Subsequently, the deformable cross-attention layers and the segmentation head bilinearly interpolate the output. Finally, the detection and segmentation losses are applied to all decoder layers during the training process (Robinson et al. 2025).

YOLO vs RF-DETR

The selection of YOLO and RF-DETR for the experimental evaluation is based on their fundamental architectural divergence. Although both models are utilized for the same purpose, object detection and localization of defects, they differ significantly in their internal working structure.

1.7. Model Training

1.8. Model Testing

1.9. Evaluation Metrics

2. Results

2.1. Model Results and Statistical Evaluation

2.2. Quantitative and Qualitative Evaluation

2.3. Evaluation Visualization

3. Discussion

3.1. Interpretation of Results

(Self Note;)Buhl (2024) sagt dass das modell weniger false positives hat, unbedingt mit dem resultat von rf detr abgleichen! ## Comparison with Expectations

4. Conclusion

4.1. Summary of Key Findings

4.2. Limitations and Challenges

4.3. Future Research Directions

5. References

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6. Lists

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7. Attachments